

A foundation model for atomistic materials chemistry

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Atomistic simulations of matter, especially those that leverage first-principles (*ab initio*) electronic structure theory, provide a microscopic view of the world, underpinning much of our understanding of chemistry and materials science. Over the last decade or so, machine-learned force fields have transformed atomistic modeling by enabling simulations of *ab initio* quality over unprecedented time and length scales. However, early ML force fields have largely been limited by: (i) the substantial computational and human effort of developing and validating potentials for each particular system of interest; and (ii) a general lack of transferability from one chemical system to the next. Here we show that it is possible to create a general-purpose atomistic ML model, trained on a public dataset of moderate size, that is capable of running stable molecular dynamics for a wide range of molecules and materials. We demonstrate the power of the MACE-MP-0 model — and its qualitative and at times quantitative accuracy — on a diverse set of problems in the physical sciences, including properties of solids, liquids, gases, chemical reactions, interfaces and even the dynamics of a small protein. The model can be applied out of the box as a starting or “foundation” model for any atomistic system of interest and, when desired, can be fine-tuned on just a handful of application-specific data points to reach *ab initio* accuracy. Establishing that a stable force-field model can cover almost all materials changes atomistic modeling in a fundamental way: experienced users get reliable results much faster, and beginners face a lower barrier to entry. Foundation models thus represent a step towards democratising the revolution in atomic-scale modeling that has been brought about by ML force fields.

1 Introduction

An overarching goal in the field of atomistic modeling is to develop an interatomic potential (alternatively also called a “force field”) that quickly and accurately predicts the total energy and atomic forces for an arbitrary chemical structure. Existing methods are not capable of this feat: while *ab initio* methods such as density functional theory (DFT) (1–7) are widely applicable and accurate, their high computational cost prohibits their use in many important cases, including high-throughput workflows and those in which large ($\gg 1000$ atom) systems need to be simulated over long timescales. Conversely, empirical force-field models that use simple functional forms are extremely cheap and so quick to use, but fail to accurately capture the important subtleties of the many-body interactions between collections of atoms induced by quantum mechanics (8). Finally, modern machine-learning based interatomic potentials (MLIPs) are capable of faithfully approximating *ab initio* methods for orders of magnitude less cost, but typically require significant upfront investment and human effort when generating and labeling the training dataset (9–17). Furthermore, these datasets and models typically need to be re-developed from scratch for each new system of interest (18). As a remedy to these issues, and as a step towards a truly universal MLIP, we present MACE-MP-0, a foundation model for materials chemistry that displays an impressive out-of-the-box ability to model a wide variety of chemical systems. Crucially, we also demonstrate that fine-tuning MACE-MP-0 using just a handful of new configurations leads to quantitatively accurate models, dramatically reducing the cost and barrier to entry for the modeling of novel chemical systems.

MACE-MP-0 uses the MACE architecture (19), which unified the atomic cluster expansion (ACE) (14, 20–22) and equivariant graph neural networks (16, 23). MACE was designed to keep only what appear to be essential components of the latter (23): the element embedding (12, 24) and the equivariant messages constructed through the symmetric tensor product operation. MACE’s unique innovations are that (i) it uses high body-order equivariant features in each layer (4-body in the present case), and consequently only two layers of message passing are sufficient; (25) (ii) it is only mildly nonlinear, as the only nonlinear activations are in the radial basis and the final readout layer, hence its classification as a graph tensor network, (iii) it uses tensor decomposition (24) for efficient parameterization of high body-order features. The MACE architecture

allows MACE-MP-0 to accurately model its training data while remaining competitively performant with other graph neural networks, presently allowing simulations of around a thousand atoms for nanoseconds per day on a GPU.

Despite training MACE-MP-0 on a dataset with a specific materials focus (MPtrj), the most striking finding is that our model shows remarkable out-of-distribution performance, and leads to stable molecular dynamics (MD) simulations for arbitrary systems over long timescales showing chemically sensible structures, reactions and transformations. In the main body of this paper, we showcase the generality of MACE-MP-0 by considering three disparate classes of chemical systems: solid and liquid water, heterogeneous catalysis, and metal-organic frameworks. In the Supplementary Information, we further demonstrate MACE-MP-0’s capabilities on an unprecedented range of qualitative and quantitative examples drawn from computational chemistry and materials science, including running molecular dynamics simulations for a wide variety of chemistries, predicting phonon spectra, calculating activation energies for point defect and dislocation motion, simulating solvent mixtures, combusting hydrogen gas, modeling a complete rechargeable battery cell, and many more.

There are several versions of the MACE-MP-0 model, all trained on the same data set, with minor variations in the model architecture. Unless otherwise stated, all results in this paper were obtained with the MACE-MP-0b3 version, and this is emphasized in figures and captions, while we retrain the simpler “MACE-MP-0” name in the text for readability. All model versions are publicly available.

2 Applications

2.1 Water and aqueous systems

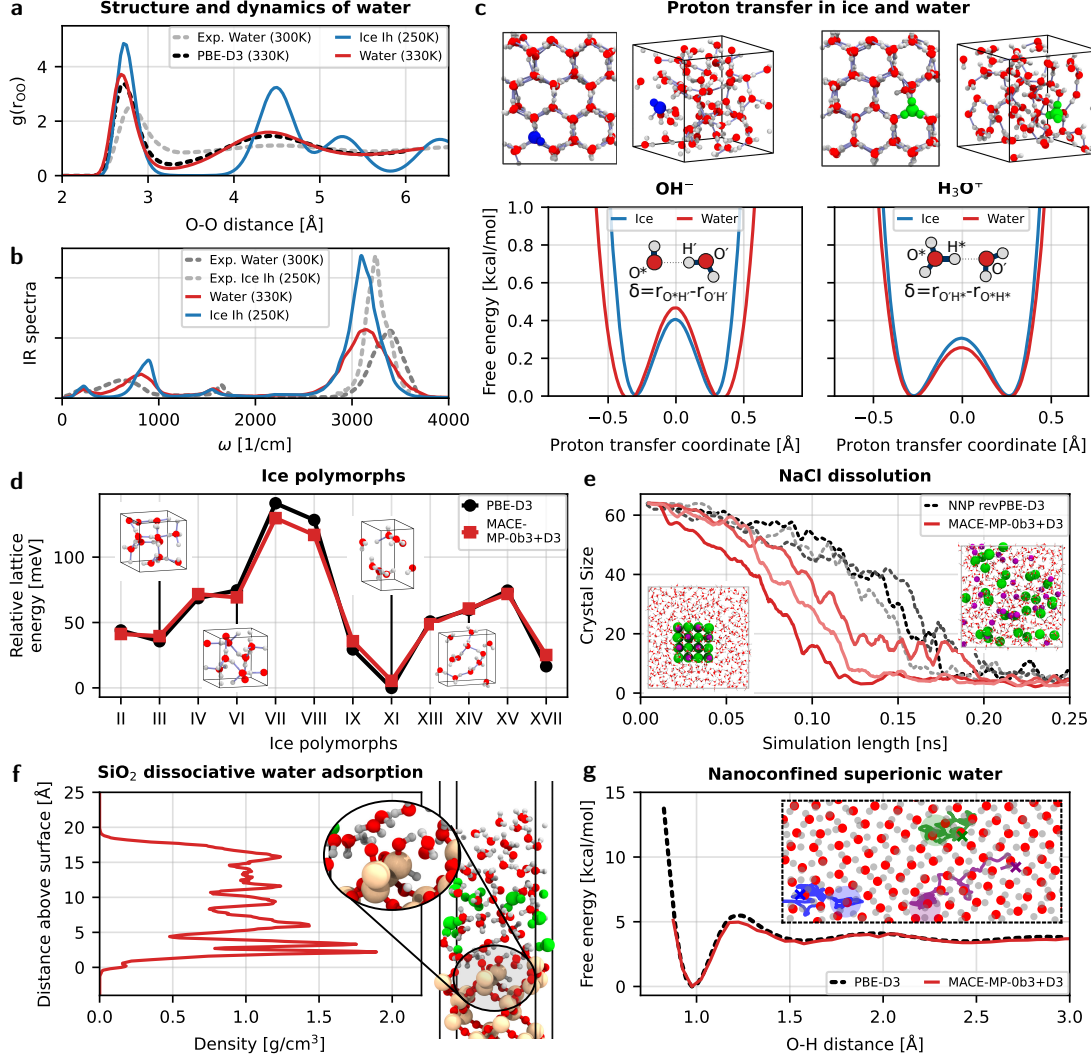


Figure 1: **Aqueous systems.** (a) Oxygen–oxygen radial distribution function for bulk water (experimental result from Ref. (26)) and ice Ih. (b) Experimental (Ref. (27, 28)) and computed infrared spectra of bulk water and ice Ih. (c) Free energy profiles as a function of the proton transfer barrier for a hydroxide ion and excess proton in ice Ih at 250 K and bulk water at 330 K. Snapshots at the top show the simulation cells. (d) Performance of MACE-MP-0b3 (red squares) on the relative lattice energies of the DMC-ICE13 dataset, compared to the reference method, PBE-D3 (29) (black circles). (e) Dissolution of a $4 \times 4 \times 4$ unit-cell NaCl nanocrystal in water at 400 K, monitoring the extent of dissolution over the simulation time via the crystal size. Performance of the MACE-MP-0b3 (red lines) is compared to a neural network potential (30) trained explicitly to capture NaCl dissolution (black dashed lines). (f) SiO_2 /water interface simulation showing density modulations and dissociative water adsorption, with an inset highlighting the deprotonation of water as indicated by a shoulder in the water density plot. H_3O^+ defects in the liquid are highlighted in green. (g) The free energy profile of the O–H distance in the superionic phase of monolayer water in a confining potential. The inset shows a snapshot of the monolayer superionic phase with lines indicating the 50 ps-long trajectory of randomly chosen hydrogen atoms with “x” indicating their initial positions.